

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
from google.colab import files
```

```
uploaded = files.upload()
```

Choose files foundation.jpeg
foundation.jpeg(image/jpeg) - 68028 bytes, last modified: 26/08/2026 - 100% done
Saving foundation.jpeg to foundation.jpeg

```
image = cv2.imread('foundation.jpeg')

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

print("Image loaded successfully!")
print("Image Shape:", gray.shape)
```

Image loaded successfully!
Image Shape: (1308, 736)

```
plt.figure(figsize=(8, 6))

plt.imshow(gray, cmap='gray')
plt.title('Original Grayscale Image')
plt.axis('off')

plt.show()
```

Original Grayscale Image



```
gaussian = cv2.GaussianBlur(gray, (5, 5), 0)

plt.figure(figsize=(8, 6))

plt.imshow(gaussian, cmap='gray')
plt.title('Gaussian Blur')
plt.axis('off')

plt.show()
```

Gaussian Blur



```
median = cv2.medianBlur(gray, 5)

plt.figure(figsize=(8, 6))

plt.imshow(median, cmap='gray')
plt.title('Median Filter')
plt.axis('off')

plt.show()
```

Median Filter



```
average = cv2.blur(gray, (5, 5))

plt.figure(figsize=(8, 6))

plt.imshow(average, cmap='gray')
plt.title('Average (Mean) Filter')
plt.axis('off')

plt.show()
```

Average (Mean) Filter



```
laplacian = cv2.Laplacian(gray, cv2.CV_64F)

laplacian = np.uint8(np.absolute(laplacian))

plt.figure(figsize=(8, 6))

plt.imshow(laplacian, cmap='gray')
plt.title('Laplacian Filter')
plt.axis('off')

plt.show()
```

Laplacian Filter



```
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)

sobel_x = np.uint8(np.absolute(sobel_x))

plt.figure(figsize=(8, 6))

plt.imshow(sobel_x, cmap='gray')
plt.title('Sobel X - Horizontal Edges')
plt.axis('off')

plt.show()
```

Sobel X - Horizontal Edges



```
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

sobel_y = np.uint8(np.absolute(sobel_y))

plt.figure(figsize=(8, 6))

plt.imshow(sobel_y, cmap='gray')
plt.title('Sobel Y - Vertical Edges')
plt.axis('off')

plt.show()
```

Sobel Y - Vertical Edges



```
sobel_combined = cv2.magnitude(
    sobel_x.astype(np.float32),
    sobel_y.astype(np.float32)
)

sobel_combined = np.uint8(sobel_combined)

plt.figure(figsize=(8, 6))

plt.imshow(sobel_combined, cmap='gray')
plt.title('Combined Sobel Edge Detection')
plt.axis('off')

plt.show()
```

Combined Sobel Edge Detection



```
plt.figure(figsize=(15, 5))

plt.subplot(1, 4, 1)
plt.imshow(gray, cmap='gray')
plt.title('Original')
plt.axis('off')

plt.subplot(1, 4, 2)
plt.imshow(gaussian, cmap='gray')
plt.title('Gaussian')
plt.axis('off')

plt.subplot(1, 4, 3)
plt.imshow(median, cmap='gray')
plt.title('Median')
plt.axis('off')

plt.subplot(1, 4, 4)
plt.imshow(average, cmap='gray')
plt.title('Average')
plt.axis('off')

plt.show()
```

Original



Gaussian



Median



Average



```
plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.imshow(laplacian, cmap='gray')
plt.title('Laplacian')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(sobel_x, cmap='gray')
plt.title('Sobel X')
plt.axis('off')
```

```
plt.subplot(1, 3, 3)
plt.imshow(sobel_y, cmap='gray')
plt.title('Sobel Y')
plt.axis('off')

plt.show()
```

Laplacian



Sobel X



Sobel Y



```
plt.figure(figsize=(15, 10))

plt.subplot(2, 3, 1)
plt.imshow(gray, cmap='gray')
plt.title('Original')
plt.axis('off')

plt.subplot(2, 3, 2)
plt.imshow(gaussian, cmap='gray')
plt.title('Gaussian Blur')
plt.axis('off')

plt.subplot(2, 3, 3)
plt.imshow(median, cmap='gray')
plt.title('Median Filter')
plt.axis('off')

plt.subplot(2, 3, 4)
plt.imshow(average, cmap='gray')
plt.title('Average Filter')
plt.axis('off')

plt.subplot(2, 3, 5)
plt.imshow(laplacian, cmap='gray')
plt.title('Laplacian')
plt.axis('off')

plt.subplot(2, 3, 6)
plt.imshow(sobel_combined, cmap='gray')
plt.title('Sobel')
plt.axis('off')

plt.tight_layout()
plt.show()
```

Original



Gaussian Blur



Median Filter



Average Filter



Laplacian



Sobel



```
print("FILTER COMPARISON")
print("-----")

print("Original Mean Intensity:", np.mean(gray))
print("Gaussian Mean Intensity:", np.mean(gaussian))
print("Median Mean Intensity:", np.mean(median))
print("Average Mean Intensity:", np.mean(average))

print("\nStandard Deviation:")
print("Original:", np.std(gray))
print("Gaussian:", np.std(gaussian))
print("Median:", np.std(median))
print("Average:", np.std(average))
```

FILTER COMPARISON

Original Mean Intensity: 179.11075447081504
 Gaussian Mean Intensity: 179.1136183270177
 Median Mean Intensity: 179.4165939535966
 Average Mean Intensity: 179.11146498138547

Standard Deviation:
 Original: 56.737555220226184
 Gaussian: 56.07669914010473
 Median: 56.083123219748295
 Average: 55.73791797214758

```
print("OBSERVATIONS")
print("-----")
```